Лабораторная работа №4

Продвинутое использование git

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Титульный слайд

Дисциплина: Архитектура компьютеров и операционные системы (раздел «Операционные системы»)
Работа: Лабораторная работа 4 — Продвинутое использование git
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Организация: Российский университет дружбы народов (РУДН) — ## Содержание 1. Актуальность и постановка задачи
2. Цель, гипотеза, задачи
3. Материалы и инструменты
4. Ход выполнения
5. Демонстрация результатов
6. Выводы и результаты — ## Информация о докладчике :::::::::::::: {.columns align=center} ::: {.column width="65%"} - **Лебедев Сергей Алексеевич** - студент направления **02.03.00 Компьютерные и**

```
lebedevsa@lebedevsa:~$ sudo dnf copr enable elegos/gitflow
[sudo] password for lebedevsa:
 https://copr.fedorainfracloud.org/api_3/rpmrepo/elegos/gitflow/fedora-43/    100% |  58.0   B
Enabling a Copr repository. Please note that this repository is not part
of the main distribution, and quality may vary.

The Fedora Project does not exercise any power over the contents of
this repository beyond the rules outlined in the Copr FAQ at
<https://docs.pagure.org/copr.copr/user_documentation.html#what-i-can-build-in-copr>,
and packages are not held to any quality or security level.

Please do not file bug reports about these packages in Fedora
Bugzilla. In case of problems, contact the owner of this repository.
Is this ok [y/N]: y
lebedevsa@lebedevsa:~$ sudo dnf install gitflow
```

Создание репозитория и подключение к GitHub

```
mkdir git-extended && cd git-extended
touch README.md
git init && git add . && git commit -m "first commit"
git remote add origin git@github.com:lebedev-s-a/git-extended.git
git push -u origin master
```

```
lebedevsa@lebedevsa:~$ mkdir git-extended
lebedevsa@lebedevsa:~$ cd git-extended
lebedevsa@lebedevsa:~/git-extended$ touch README.md
lebedevsa@lebedevsa:~/git-extended$ git add .
fatal: not a git repository (or any parent up to mount point /)
Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).
lebedevsa@lebedevsa:~/git-extended$ git init
Initialized empty Git repository in /home/lebedevsa/git-extended/.git/
lebedevsa@lebedevsa:~/git-extended$ ls
README.md
lebedevsa@lebedevsa:~/git-extended$ git add .
lebedevsa@lebedevsa:~/git-extended$ git commit -m "first commit"
[master (root-commit) 117f6bc] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
```

```
lebedevsa@lebedevsa:~/git-extended$ git remote add origin git@github.com:lebedev-s-a/get-extended.git
lebedevsa@lebedevsa:~/git-extended$ git push -u origin master
ERROR: Repository not found.
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
lebedevsa@lebedevsa:~/git-extended$ git remote remove origin
lebedevsa@lebedevsa:~/git-extended$ git remote add origin git@github.com:lebedev-s-a/git-extended.git
lebedevsa@lebedevsa:~/git-extended$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 868 bytes | 868.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:lebedev-s-a/git-extended.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Настройка Conventional Commits

Инициализация проекта Node.js и добавление конфигурации commitizen в `package.json`:

```
"config": {
  "commitizen": {
    "path": "cz-conventional-changelog"
  }
}
```

Коммит через интерактивное меню:

```
git cz
```

```
lebedevsa@lebedevsa:~/git-extended$ nano package.json
lebedevsa@lebedevsa:~/git-extended$ git add .
lebedevsa@lebedevsa:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: docs:     Documentation only changes
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 94 chars):
 (1) 0
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master f25f5fd] docs: 0
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
lebedevsa@lebedevsa:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
   - master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/lebedevsa/git-extended/.git/hooks]
```

Инициализация git-flow

```
git flow init
```

Настраиваются ветки: `master`, `develop`, `feature/`, `release/`, `hotfix/`
Префикс для ярлыков версий: **v**

```
lebedevsa@lebedevsa:~/git-extended$ git branch
* develop
  master
lebedevsa@lebedevsa:~/git-extended$ git push --all
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.14 KiB | 1.14 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:lebedev-s-a/git-extended.git
   117f6bc..f25f5fd  master -> master
 * [new branch]      develop -> develop
lebedevsa@lebedevsa:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
lebedevsa@lebedevsa:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

     git flow release finish '1.0.0'

lebedevsa@lebedevsa:~/git-extended$ standard-changelog --first-release
lebedevsa@lebedevsa:~/git-extended$ git add CHANGELOG.md
lebedevsa@lebedevsa:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 8d453b8] chore(site): add changelog
```

Создание релиза v1.0.0 и CHANGELOG

```
git flow release start 1.0.0
standard-changelog --first-release
git add CHANGELOG.md
git commit -am "chore(site): add changelog"
git flow release finish 1.0.0
```

```
lebedevsa@lebedevsa:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.68 KiB | 2.68 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:lebedev-s-a/git-extended.git
   f25f5fd..81ccccf  develop -> develop
   f25f5fd..0df3fe9  master -> master
lebedevsa@lebedevsa:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 160 bytes | 160.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:lebedev-s-a/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
lebedevsa@lebedevsa:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/lebedev-s-a/git-extended/releases/tag/v1.0.0
```

Работа с feature-веткой и релиз v1.2.3

```
git flow feature start feature_branch
# ... разработка ...
git flow feature finish feature_branch
git flow release start 1.2.3
standard-changelog
git add CHANGELOG.md
git commit -am "chore(site): update changelog"
git flow release finish 1.2.3
```

```
lebedevsa@lebedevsa:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Updating 81ccccf..324459d
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
Deleted branch feature/feature_branch (was 324459d).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

lebedevsa@lebedevsa:~/git-extended$ git flow release start 1.2.3
Branches 'develop' and 'origin/develop' have diverged.
And local branch 'develop' is ahead of 'origin/develop'.
Switched to a new branch 'release/1.2.3'

Summary of actions:
- A new branch 'release/1.2.3' was created, based on 'develop'
- You are now on branch 'release/1.2.3'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.2.3'
```

Публикация на GitHub

Отправка всех веток и тегов:

```
git push --all
git push --tags
```

Создание релизов на GitHub:

```
gh release create v1.0.0 -F CHANGELOG.md
gh release create v1.2.3 -F CHANGELOG.md
```

```
lebedevsa@lebedevsa:~/git-extended$ git push --all
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 4 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 3.81 KiB | 3.81 MiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:lebedev-s-a/git-extended.git
```

```
lebedevsa@lebedevsa:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 160 bytes | 160.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:lebedev-s-a/git-extended.git
 * [new tag]         v1.2.3 -> v1.2.3
lebedevsa@lebedevsa:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/lebedev-s-a/git-extended/releases/tag/v1.2.3
lebedevsa@lebedevsa:~/git-extended$
```

Итог и выводы

Результаты: - Установлены и настроены git-flow, commitizen, standard-changelog - Создан тестовый репозиторий с подключением к GitHub - Настроены Conventional Commits через `package.json` - Выполнена работа с ветками feature и release в рамках Gitflow - Сгенерирован `CHANGELOG.md` и опубликованы релизы v1.0.0 и v1.2.3 на GitHub

Вывод:
получены практические навыки организации профессионального рабочего процесса с git на основе Gitflow Workflow и Conventional Commits — ## Ресурсы - Gitflow: https://nvie.com/posts/a-successful-git-branching-model/ - git-flow (Copr): https://copr.fedorainfracloud.org/coprs/elegos/gitflow/ - Conventional Commits: https://www.conventionalcommits.org/ - Semantic Versioning: https://semver.org/ - Conventional Changelog: